

Guido Mazzuca, PhD

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EDUCATION

SISSA, Trieste, Italy
PhD Mathematics with honors
Thesis: *Nonlinear Lattices and Random Matrices*
Advisors: Tamara Grava and Alberto Maspero.

University of Milan, Italy
MSc. in Mathematics with honors

University of Milan, Italy
BSc. in Mathematics

RESEARCH INTERESTS

My research focuses on integrable systems, spanning both continuous partial differential equations (such as NLS, KdV, and mKdV) and discrete lattice models (including Toda and Ablowitz-Ladik systems). I investigate the long-time behavior and dispersive properties of solutions, with a particular interest in the dynamics of soliton gases. A significant part of my work explores the deep connections between these integrable systems and Random Matrix Theory, primarily through the application of the Riemann-Hilbert problem approach, both in its standard and random formulations.

RESEARCH EXPERIENCE

Postdoctoral researcher, Tulane University
New Orleans, Louisiana, USA.
Mentor: Professor Kenneth McLaughlin

08/11/2023 – Present

- Deduced the behavior of correlation functions for the Volterra lattice by applying the theory of Generalized Hydrodynamics.
- Investigated the behavior of the NLS with random initial data, collaborating with Profs. E. Girotti, R. Jenkins, M. Yattselev, T. Grava, and K. McLaughlin.
- Analyzed the behavior of the NLS equation in the *condensed soliton gas* regime in joint work with Prof. K. McLaughlin and Dr. A. Gkougkou.
- Derived a Large Deviation principle for the Jacobi-Muttalib-Borodin ensemble and computed its density of states in collaboration with Prof. A. Occelli and Prof. J. Husson.
- Characterized the combinatorial and analytical properties of the density of states of the q -Laguerre polynomials alongside Prof. Sung-Soo Byun and Yeong-Gwang Jung.
- Discovered the universal behavior of the N -soliton solution with maximal modulus in joint work with Prof. K. McLaughlin and Dr. A. Gkougkou.

- Spearheaded a collaboration with Dr. G. Orsatti and Dr. A. Gkogkou to study the long-time behavior of the NLS soliton gas.
- Authored five papers resulting from this research—three published and two submitted—with two additional manuscripts in preparation.

Junior Fellow, Mittag-Leffler Institute
Stockholm, Sweden

09/01/2024 – 12/13/2024

- Fostered a collaboration with Dr. M. Piorkowski at KTH to compute the correlation functions of the infinite Toda lattice with random initial data.
- Established a joint project with Dr. D. Ofner at KTH to define a new exactly solvable random matrix ensemble related to the *Itoh–Narita–Bogoyavlenskii* lattice.
- Secured funding for this position through the Junior Fellowship awarded by the Mittag-Leffler Institute, advancing two manuscripts currently in preparation.

Postdoctoral researcher, KTH
Stockholm, Sweden

10/01/2021 – 08/10/2023

Mentor: Professor Håkan Hedenmalm

- Executed a deep numerical investigation of the correlation functions of the Toda lattice, demonstrating an astonishing agreement with Generalized Hydrodynamic predictions in collaboration with Profs. K. McLaughlin, T. Kriecherbauer, H. Spohn, T. Grava, and C. Mendl.
- Uncovered a connection between the Circular β ensemble and the Ablowitz-Ladik lattice, enabling the computation of moments and the density of states for the lattice.
- Formulated connections between several random matrix ensembles and integrable systems to compute the density of states for these models alongside Prof. T. Grava, Dr. M. Gissoni, and Prof. G. Gubbiotti.
- Established a central limit theorem-type result for all classical β ensembles at high temperature and several integrable models in joint work with Dr. R. Memin.
- Published four peer-reviewed articles funded by the KAW grant 2018.0344.

Program Associate, SLMath (ex MSRI)
Berkeley, California, USA

08/16/2021 – 09/15/2021

- Contributed to the program “Universality and Integrability in Random Matrix Theory and Interacting Particle Systems.”
- Derived a large deviation principle for the Ablowitz-Ladik lattice and for the Circular β ensemble at high temperature in collaboration with Dr. R. Memin.
- Published one paper based on this research, partially funded by the IPaDEGAN grant and the Simons Laufer Mathematical Sciences Institute.

Graduate Student, SISSA
Trieste, Italy

10/01/2017 – 09/24/2021

Advisor: Professor Tamara Grava and Professor Alberto Maspero

- Calculated the density of states and the moments of the classical β ensemble at high temperature using loop equations, and defined the anti-symmetric Gaussian α ensemble in joint work with Prof. P. J. Forrester.
- Computed the correlation functions for the short-range harmonic chain, asymptotically describing the behavior of their peaks in terms of special functions with Profs. K. McLaughlin, T. Grava, and T. Kriecherbauer.

- Identified a relationship between the classical β ensemble at high temperature and specific integrable models to compute their density of states in particular regimes.
- Demonstrated that the conserved quantities of the Toda lattice serve as adiabatic invariants for the Fermi-Pasta-Ulam-Tsingou model, explaining its almost periodic behavior.
- Produced four publications resulting from this research, partially funded by the IPaDEGAN grant.

PUBLICATIONS

Published papers:

1. S. Byun, Y. Jung, G. Mazzuca, *q-deformation of the Marchenko–Pastur law*, *International Mathematics Research Notices*, Volume 2026, Issue 13, July 2026, rnag134, DOI:10.1093/imrn/rnag134
2. M. Girotti, T. Grava, R. Jenkins, G. Mazzuca, K. McLaughlin, M. Yattselev: *Soliton Synchronization with Randomness: Rogue Waves and Universality*. *Nonlinearity* DOI: 10.1088/1361-6544/ae1435.
3. A. Gkogkou, G. Mazzuca, K. McLaughlin. *The formation of a soliton gas condensate for the focusing nonlinear Schrödinger equation*. *Journal of Nonlinear Waves*. 2025;1:e14. doi:10.1017/jnw.2025.10016
4. G. Mazzuca, *Generalized Hydrodynamics for the Volterra lattice: ballistic and nonballistic behavior of correlation functions*. *Journal of Physics A: Mathematical and Theoretical*. DOI: 10.1088/1751-8121/ad742b
5. G. Mazzuca, R. Memin: *CLT for β ensembles at high-temperature, and for integrable systems: a transfer operator approach* *Ann. Henri Poincaré* (2024). DOI: 10.1007/s00023-024-01435-0
6. G. Mazzuca, T. Grava, T. Kriecherbauer, K. T-R McLaughlin, C. B. Mendl, H. Spohn: *Equilibrium Spacetime Correlations of the Toda Lattice on the Hydrodynamic Scale*, *Journal of Statistical Physical Physics*, DOI: <https://doi.org/10.1007/s10955-023-03155-x>
7. G. Mazzuca, and T. Grava: *Generalized Gibbs ensemble of the Ablowitz-Ladik lattice, circular β -ensemble and double confluent Heun equation*. *Communication in Mathematical Physics*, DOI: 10.1007/s00220-023-04642-8 (2023).
8. M. Gisonni, T. Grava, G. Gubbiotti, G. Mazzuca: *Discrete integrable systems and random Lax matrices*. *Journal of Statistical Physics* 190, Article number: 10 (2023). DOI: 10.1007/s10955-022-03024-z
9. G. Mazzuca, and R. Memin: *Large Deviations for Ablowitz-Ladik lattice, and the Schur flow*. *Electronic Journal of Probability*. DOI: 10.1214/23-EJP941.(2023)
10. G. Mazzuca: *On the mean Density of States of some matrices related to the beta ensembles and an application to the Toda lattice*. *Journal of Mathematical Physics* 63, 043501 (2022). DOI: 10.1063/5.0076539.
11. G. Mazzuca, and P.J. Forrester: *The classical beta ensembles with beta proportional to 1/N: from loop equations to Dyson’s disordered chain*. *Journal of Mathematical Physics*, Special Issue in honour of F. Dyson, 2021. DOI: 10.1063/5.0048481.
12. T. Grava, T. Kriecherbauer, G. Mazzuca, and K. D. T.-R. McLaughlin: *Correlation functions for a chain of short range oscillators*. *Journal of Statistical Physics* (2021). DOI: 10.1007/s10955-021-02735-z
13. T. Grava, A. Maspero, G. Mazzuca, and A. Ponno: *Adiabatic invariants for the FPUT and Toda chain in the thermodynamic limit*. *Communications in Mathematical Physics*, 380 (2020), pp. 811-851. DOI: 10.1007/s00220-020-03866-2.

Preprint :

1. J. Husson, G. Mazzuca, A. Occelli: *Discrete and Continuous Muttalib–Borodin process: Large deviations and Riemann–Hilbert analysis*. ArXiv preprint arXiv:2505.23164
2. A. Gkogkou, G. Mazzuca, K. D. T-R McLaughlin: *Painlevé Universality classes for the maximal amplitude solution of the Focusing Nonlinear Schrödinger Equation with randomness*. ArXiv preprint arXiv: 2602.05101

RESEARCH GRANTS

Contribution to prior funded research

IPaDEGAN

2018–2023

- Served as a contributor during PhD, developing new approaches to study integrable dynamical systems with randomness.
- Principal Investigator: Professor Gregorio Falqui
- Funded Under: Marie Skłodowska-Curie Actions (MSCA RISE 778010), 886500€ (\$ 970000).

Personal Grant

MATRIX-Simon Travel Grant

2025

- Awarded \$2000 (2500 AUD) by the MATRIX institute in Australia to participate in the research program “Log-gases in Caeli Australi: Recent Developments in and around Random Matrix Theory” in Creswick.

Postdoc Travel Grant

2024–2025

- Received \$2500 (3 different awards over 2 years) from the Office of Graduate and Postdoctoral Studies of Tulane University to present research at conferences in Europe and the USA.

Research grant

2019

- Granted 2000€ (\$ 2200) by LYSM to attend the semester “Integrability and Randomness in Mathematical Physics” at CIRM.

FELLOWSHIPS

Junior Fellowship

Fall 2024

- Awarded 80000 SEK (\$ 7500) and housing by the Mittag-Leffler Institute, Stockholm, Sweden, for postdoctoral research.

Marie-Curie Fellowship

2023

- Awarded 200000€ (\$ 220000) by the European Research Council as Principal Investigator (Declined for personal reasons).

PhD Fellowship

2017–2021

- Received a 57000€ (\$ 62500) doctoral fellowship from SISSA (International School for Advanced Studies).

HONORS & AWARDS

Lutman Prize 2022
Best PhD thesis in Mathematics, 1000€(\$ 1100)

SERVICE

Applied Math Seminar 2026–2027
Tulane University, New Orleans, LA, USA
Role: Organizer

AMS Sectional Meeting: Integrability and Beyond Fall 2026
Arizona State University, Tempe, AZ, USA
Role: Organizer

AMS Sectional Meeting: Integrability and Randomness, Fall 2025
Tulane University, New Orleans, LA, USA
Role: Organizer

Summer School: “Mini Math School” 2025–2024
Scope: Summer School for PhD students of Tulane Mathematics department
Tulane University, New Orleans, LA, USA
Role: Creator and Organizer

Conference: Integrable Systems in Geometry and Mathematical Physics 2021
Conference in memory of Boris Dubrovin
SISSA, Trieste, Italy
Role: Co-organizer

Workshop: Integrable Systems Around the World 2020
Online Workshop
Role: Co-organizer

Conference: Integrability and Randomness in Mathematical Physics and Geometry 2019
CIRM, Marseilles, France
Role: Co-organizer

Research School: Coulomb Gas, Integrability and Painlevé Equations 2019
CIRM, Marseilles, France
Role: Co-organizer

SELECTED TALKS

For a complete list of my talk, you can visit my personal web-page.

“Universality classes for the NLS rogue waves” July 2026
Institut Henri Poincaré, Paris, France.

“CLT for the real β ensemble and the Toda lattice” September 2024
Institute Mittag-Leffler, Program Seminar, Stockholm, Sweden

“Ablovitz-Ladik lattice and the Circular β -ensemble: a surprising connection” June 2024
Conference: 4th Italian Meeting on Probability and Mathematical Statistics, Rome, Italy

“LDP for the Ablovitz-Ladik lattice and the circular β ensemble at high-temperature” May 2023
KTH departmental seminar, Stockholm, Sweden

“From the Ablowitz-Ladik lattice to the Circular beta-ensemble” MEGA seminar series, Lyon, France	May 2022
“Alpha ensembles and integrable systems” Mathematical Science Research Institute (MSRI)	September 2021
“Adiabatic Invariants for Toda and FPUT” Workshop “Integrability and Nonlinear Dispersive Equations” CIRM Marseille, France.	June 2019

TEACHING AND MENTORING EXPERIENCE

Co-advisor	Fall 2025 – Present
I am currently co-advising a graduate student on a project regarding tilings of the Aztec diamond and Orthogonal Polynomials.	

Instructor	Fall 2026
Introduction to Applied Math, Tulane university. 200-level course.	

Mentor	Spring 2026
Senior Seminar Project: I supervised an undergraduate student on a project regarding modeling of reaction time.	

Instructor	Spring 2026
Complex Analysis, Tulane university. 400/600-level course; offered both to undergraduate and graduate students.	

Instructor	Fall 2025
Calculus III, Tulane university. 200-level course.	

Instructor and coordinator	Spring 2024 and Spring 2025
Calculus II, Tulane university. 100-level course.	

Instructor	Fall 2023
Calculus I, Tulane university. 100-level course.	

Tutor	May 2022
Graduate Course “Integrable systems with random initial data” LMS-CMI Research School “Methods for Random Matrix Theory and Applications” Reading, UK	

VOLUNTEER WORK

GiST (Girls in Stem at Tulane) and BATS	Fall and Spring 2025’2026
I participated in a series of events designed for high school students.	

SISSA 4 School	2017–2021
I participated in a series of events designed for high school students.	

Trieste NEXT	2017–2019
I participated in this scientific fair, designing activities for high-school and middle school students.	